UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Data representation and number systems

Convert 286610 to octal. Show your complete workflow.

Answer: 286610 = 54628

- *1 mark for identifying the target base.*
- *1 mark for a correct repeated-division or place-value method.*
- *1 mark for the correct digit sequence.*
- *1 mark for the correct base subscript.*
- *1 mark for a complete, consistent workflow.*

QUESTION A2 (5 marks)

Topic: Computer generations and history

The evolution of computers and technological advancements have led to powerful variations that can process data faster, perform complex calculations, support artificial intelligence, enable global communication, manage vast databases, simulate real-world scenarios, and offer automation, all while being smaller, more energy-efficient, and widely accessible.
a) Name the primary technology in the zero, first, and second computer generations.
b) List TWO (2) fundamental components of integrated circuits.

Source: CSC429 (18).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Parallel and multiprocessor architecture

a)
List THREE (3) characteristics of Symmetric Multiprocessor (SMP).
b)
Describe ONE (1) advantage of Symmetric Multiprocessor (SMP) interms of availability.

Source: CSC429 (13).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Memory hierarchy and cache

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a) Discuss THREE (3) main key factors for memory selection.
b) List TWO (2) examples for internal cabling memory and TWO (2) examples of system memory.

Source: CSC429 (6).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Input/output and interrupts

a) Why must the CPU be faster than all the other inpuuoutput devices?
b) How do the I/O controllers facilitate communication between the CPU and I/O devices?
c) Discuss TWO (2) types of I/O modules.
d) When would an I/O controller of a specific device be installed?

Source: CSC429 (20).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Computer architecture and CPU organisation

a) Draw a complete machine cycle by giving the exact location of each phase.
b) Briefly, describe each phase in the machine cycle.

Source: CSC429 (8).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

Processor contains a number of registers used for temporary storage of data. Describe FIVE (5) types of registers.

Source: CSC429 (2).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Input/output and interrupts

a) Describe THREE (3) types of interfaces to external devices.
b) Explain TWO (2) types of interrupt.

Source: CSC429 (10).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.
a) Convert 156510 to hexadecimal.
b) Convert BDC16 to decimal.

a) Convert 156510 to hexadecimal.

Answer: 156510 = 61D16

- *1 mark for repeated division by 16.*
- *1 mark for correct remainders.*
- *1 mark for reading remainders in reverse order.*
- *1 mark for the correct hexadecimal digits.*
- *1 mark for the base subscript.*

b) Convert BDC16 to decimal.

Answer: BDC16 = 303610

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- *1 mark for place values 16^2, 16^1 and 16^0.*
- *1 mark for correct digit values.*
- *1 mark for correct products.*
- *1 mark for correct addition.*
- *1 mark for the final decimal value.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 5 and Table 1
z-A+c2/B*D-3A
Figure 5
Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 5 using instructions for zero-address, one-address, and three-address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: lt/achine instruction
Three-address
Zero-address
One-address
ADD (p,q,r)
LOAD p
PUSH p
SUB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

Source: CSC429 (21).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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